

# A COMPUTED BOUNDING COCHAIN: THE PRETZEL KNOT $P(-2, 3, 5)$

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**ABSTRACT.** In the pillowcase (symplectic) side of the knot Atiyah–Floer conjecture, the naive Lagrangian–Floer homology of a tangle decomposition is not a knot invariant; the conjectural repair, due to Cazassus–Herald–Kirk–Kotelskiy, is a *bounding cochain*  $b$  supported on the self-intersections of the immersed curves and satisfying a mod-2 Maurer–Cartan equation. Smith proved that for the pretzel knot  $P(-2, 3, 5) = T(3, 5)$  the bounding cochain of one tangle must be nonzero, but produced no value, and to date no nonzero pillowcase bounding cochain has been computed. Working entirely within the immersed-curve combinatorial model of Herald–Kirk and Smith, we reconstruct the two Lagrangians of Smith’s decomposition of  $P(-2, 3, 5)$  from the quaternionic representation theory, reproducing his count of nine generators, two immersed bigons, and rank-5 Floer homology as a validation gate, and then compute the bounding cochain explicitly:

$$b_2 = s_A + s_B,$$

the sum of two self-intersection points of the perturbed Conway-sum Lagrangian  $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$ , one on each pillowcase seam. It is the unique minimal bounding cochain for which the deformed Floer homology has rank  $7 = \text{rank } I^{\natural}(P(-2, 3, 5))$ . The mechanism is a single immersed quadrilateral (a higher  $A_{\infty}$  product  $\mu^3$ ) whose two interior vertices are  $s_A, s_B$  and which cancels one of the two bigons modulo 2; the Maurer–Cartan equation is satisfied automatically because the relevant self-Floer products of the Lagrangian all vanish. The value is stable under change of perturbation. This is, to our knowledge, the first explicitly computed nonzero bounding cochain in pillowcase Floer theory. We are careful to separate what is proved, what is computed within the model, and what remains conjectural (the faithfulness of the model).

## 1. INTRODUCTION

**1.1. The pillowcase, and why a bounding cochain is needed.** Let  $K \subset S^3$  be a knot presented as a union of two tangles,  $K = (B^3, T_1) \cup_{(S^2, 4)} (B^3, T_2)$ , glued along a four-punctured sphere. The traceless  $\text{SU}(2)$  character variety of  $(S^2, 4)$  is the *pillowcase*

$$P = (\mathbb{R}/2\pi\mathbb{Z})^2 / \iota, \quad \iota(\gamma, \theta) = (-\gamma, -\theta),$$

a two-sphere with four  $\mathbb{Z}/2$  orbifold points (the *corners*) at  $(\gamma, \theta) \in \{0, \pi\}^2$  and symplectic form descending from  $d\gamma \wedge d\theta$  on the smooth locus  $P^*$  [8, 3]. Each tangle  $T_i$  determines an immersed Lagrangian  $\mathcal{R}_{\pi_i}(T_i) \looparrowright P^*$  (its perturbed traceless character variety), and the intersection points  $\mathcal{R}_{\pi_1}(T_1) \cap \mathcal{R}_{\pi_2}(T_2)$  are the traceless representations of  $\pi_1(S^3 \setminus K)$ —the generators of the reduced singular instanton chain complex of Kronheimer–Mrowka [6, 7]. Hedden, Herald and Kirk [3, 4] proposed building a knot invariant as the Lagrangian–Floer homology  $HF(\mathcal{R}_{\pi_1}(T_1), \mathcal{R}_{\pi_2}(T_2))$  of this pair, an Atiyah–Floer counterpart of  $I^{\natural}(K)$ .

This naive homology is *not* a knot invariant: it changes under the tangle moves that fix  $K$ , because of figure-eight bubbling at the immersed curves’ self-intersections. Cazassus, Herald, Kirk and Kotelskiy [1] identified the repair. To each tangle one attaches a *bounding cochain*  $b \in CF(\mathcal{R}(T), \mathcal{R}(T))$ —a formal  $\mathbb{F}_2$ -combination of the self-intersection points—subject to the

Maurer–Cartan equation

$$\sum_{k \geq 0} \mu^k(b, \dots, b) = 0, \quad (1)$$

where  $\mu^k$  counts immersed  $(k+1)$ -gons and  $\mu^1$  is the Floer differential. The bounding cochain deforms the differential of  $CF(\mathcal{R}(T_1), \mathcal{R}(T_2))$  into  $\partial_b$ , whose homology  $HF((\mathcal{R}(T_1), b_1), (\mathcal{R}(T_2), b_2))$  is conjecturally the knot invariant and equal to  $I^\natural(K)$  [1, Conj. 1.1].

**1.2. The problem and the result.** The bounding cochain is the analytic heart of the theory, and the hardest part to make explicit. It is inert precisely where the curves are embedded: for two-bridge knots the Lagrangians are straight arcs with no self-intersections,  $b = 0$ , and the naive homology is already  $I^\natural$  [3, 4]. Off that family it is genuinely nonzero. Two facts are known. For the torus knot  $T(4, 5)$ , Cazassus–Herald–Kirk–Kotelskiy exhibit a bounding cochain on the *earring* tangle that corrects the rank from 9 to the true 7 [1, §10]. For the pretzel knot  $P(-2, 3, 5)$ , Smith [9, Thm. 5.9] proves that the bounding cochain  $b_2$  of the summand tangle  $T_2 = Q_{1/3} + Q_{1/5}$  must be nonzero—by computing that the naive homology has rank 5 while  $I^\natural = 7$ —but he stops there: no value, and no algorithm, is given. In both cases the underlying combinatorial data was, until now, the extent of what had been extracted; *no nonzero pillowcase bounding cochain has been computed*.

We compute Smith’s  $b_2$ . Our main result is the following. Throughout, “computed within the model” means computed in the immersed-curve combinatorial Floer theory of [5, 9], in which Lagrangian–Floer polygon counts are winding-number computations for piecewise-linear immersed curves in  $P$ ; the faithfulness of that model is the standing conjecture of the subject (Section 5).

**Theorem 1.1.** *Consider  $P(-2, 3, 5) = T(3, 5)$  with Smith’s decomposition  $T_1 = \widehat{Q_{-1/2}}$  (with an earring) and  $T_2 = Q_{1/3} + Q_{1/5}$ , and let  $\mathcal{R}^\natural(T_1)$  (red) and  $\mathcal{R}_t(T_2)$  (blue) be the two immersed pillowcase Lagrangians. Within the combinatorial model:*

- (i) *Red and blue have nine intersection points and exactly two immersed bigons with distinct vertices, so  $\text{rank } HF((\mathcal{R}^\natural(T_1), 0), (\mathcal{R}_t(T_2), 0)) = 5$  (Smith’s gate, reproduced).*
- (ii) *The blue Lagrangian has forty self-intersection points, and its self-Floer products  $\mu^0, \mu^1, \mu^2$  all vanish; consequently every  $\mathbb{F}_2$ -combination of at most two self-intersections satisfies the Maurer–Cartan equation (1).*
- (iii) *There is a unique bounding cochain of minimal support that raises the rank to  $I^\natural$ , namely*

$$b_2 = s_A + s_B$$

*the sum of the two self-intersections of blue lying nearest the coordinates  $s_A \approx (\gamma, \theta) = (0.03, 1.27)$  and  $s_B \approx (3.06, 4.98)$ , one on each seam  $\gamma \in \{0, \pi\}$ ; the deformed complex then has*

$$\text{rank } HF((\mathcal{R}^\natural(T_1), 0), (\mathcal{R}_t(T_2), b_2)) = 7 = \text{rank } I^\natural(P(-2, 3, 5)).$$

The mechanism is clean and is described in Section 4: turning on  $b_2$  activates a single immersed *quadrilateral* (a term of the higher product  $\mu^3$ ) with interior vertices  $s_A, s_B$  and boundary vertices the two generators of one of the two bigons; over  $\mathbb{F}_2$  this quadrilateral cancels that bigon, dropping  $\text{rank } \partial_b$  from 2 to 1 and so raising the homology rank from 5 to 7. The two support crossings carry no other polygons at all, which is exactly why the cancellation is surgical and why  $s_A + s_B$ —rather than some larger combination—is the answer.

**1.3. On the direction of the correction.** It is worth contrasting  $P(-2, 3, 5)$  with the only prior example. For  $T(4, 5)$  the bounding cochain lives on the earring tangle and *creates* a differential, lowering the rank from 9 to 7 [1]; the deformed differential gains a term. For  $P(-2, 3, 5)$  the naive rank is 5, below the target 7, so the bounding cochain must *destroy* a differential—cancel one of the two existing bigons—raising the rank. This is the opposite, and combinatorially the more delicate, direction: a canceling polygon must land on precisely the chain-level entry of an existing bigon. That it does so, through a quadrilateral rather than a triangle, is the content of the computation.

**1.4. Method and honesty.** We did not take the immersed curves from [9] as input. Both Lagrangians were reconstructed from first principles—the quaternionic representation theory of the tangles, Smith’s fiber-product description of the Conway sum, his cut-and-paste resolution of the seam circles, and the Herald–Kirk earring—and only then validated against the numerical output of [9]: the convention derivations, the seam fiber-circle arc, the resolution of the worked example  $\mathcal{R}_t(Q_{1/2} + Q_{-1/3})$ , and finally the nine generators, two bigons and rank 5 of Theorem 1.1(i) all reproduce Smith’s published figures and numbers. Only after that gate did we compute the bounding cochain, which Smith does not.

We are explicit about the three registers, as in the companion paper [10]. The construction of the Lagrangians and the deformed-differential formalism are (i) reproduced from [3, 4, 1, 5, 9], independently re-derived and re-implemented here. The polygon and Maurer–Cartan counts, the rank computation, and the identification of  $b_2$  are (ii) computer computations, carried out by a short self-contained program included with the source and validated against [9] at every prior stage; we report them as such. The value of  $b_2$  is the value of Smith’s nonzero bounding cochain (iii) *under* the standing conjecture that the combinatorial model computes the Floer-theoretic invariant—the same hypothesis under which every computation in this subject, including [1, 9], is stated. We prove no new theorem about the analytic Atiyah–Floer correspondence.

The paper is organized as follows. Section 2 recalls the pillowcase and assembles the two Lagrangians, recording the validation gate. Section 3 states the combinatorial Floer package: winding-number bigon counting, the lift bookkeeping on the double cover, and the deformed differential. Section 4 is the computation: the Maurer–Cartan vanishing, the rank condition, the canceling quadrilateral, and the proof of Theorem 1.1. Section 5 discusses robustness, reproducibility, and the standing conjecture.

**Acknowledgements.** The immersed-curve model and the geometry we reconstruct are due to Hedden–Herald–Kirk [3, 4], Cazassus–Herald–Kirk–Kotelskiy [1], Herald–Kirk [5], and Smith [9]; we have tried to attribute each ingredient precisely.

## 2. THE TWO LAGRANGIANS OF $P(-2, 3, 5)$

**2.1. The pillowcase and immersed curves.** We use  $\mathrm{SU}(2)$  as the unit quaternions; a traceless element is a purely imaginary unit, i.e. a point of the 2-sphere  $S \subset \mathrm{Im} \mathbb{H}$ . The traceless character variety of the four-punctured sphere is the pillowcase  $P = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$  of the introduction, coordinatized by  $(\gamma, \theta)$  with  $\iota(\gamma, \theta) = (-\gamma, -\theta)$  and corners  $P^c = \{0, \pi\}^2$  [8, 3]. It is convenient to work on the double cover  $T^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$ , where  $\iota$  is the elliptic involution and a curve in  $P$  lifts to an  $\iota$ -invariant family; pillowcase counts are recovered from the cover as explained in Section 3.

A rational tangle  $Q_r$  has for its traceless character variety a straight line of slope  $r$  through the corners [3]; the torus-knot tangle  $T(3, n)$  has a non-abelian arc [3, 10]. Two operations build the curves we need:

- *Conway sum*  $T_1 + T_2$ . On the pillowcase the sum is a fiber product over the seam coordinate  $\gamma$ : the  $\theta$ -coordinates add,  $[\gamma, \theta_1 + \theta_2]$ , and over each seam  $\gamma \in \{0, \pi\}$  a circle of reducibles (a *seam fiber circle*) appears when the two summands are both non-abelian there [9, Thm. 4.22]. Perturbing the middle piece resolves each circle by a cut-and-paste that reconnects the four incident branches diagonally [9, Thm. 4.23], producing one new self-intersection per resolved circle.
- *Earring*  $\widehat{Q_r}$ . Adding the earring tangle replaces the arc of a tangle by its Herald–Kirk figure-eight  $\mathcal{R}^\natural$ : the doubled arc closing through the corner folds, with one forced self-crossing (the *pinch*) [5, Def. 6.6]. For a rational tangle the pinch supports the trivial bounding cochain [9, Lem. 5.6].

**2.2. Assembling red and blue.** Following Smith [9, §5.3], decompose  $K = P(-2, 3, 5)$  as  $T_1 = \widehat{Q_{-1/2}}$  (to which the earring is added, so that its bounding cochain vanishes) and  $T_2 = Q_{1/3} + Q_{1/5}$ . The two Lagrangians are:

- **red**  $= \mathcal{R}^\natural(T_1)$ , the earring figure-eight of the line of  $Q_{-1/2}$  in the pairing frame. It is a single immersed circle in  $P$  with one self-intersection (the pinch), on which the bounding cochain is 0.
- **blue**  $= \mathcal{R}_t(T_2) = \mathcal{R}_t(Q_{1/3} + Q_{1/5})$ , the perturbed Conway sum. It has two seams, each carrying two fiber circles; the four resolutions make it a single immersed circle in  $P$  with *forty* self-intersection points.

We built both curves as explicit piecewise-linear  $\iota$ -invariant families on  $T^2$  from the representation theory—the derivation of the compass convention and the Conway-sum fiber coordinate, the seam fiber-circle arc  $[|\theta_1 - \theta_2|, \pi - |\pi - \theta_1 - \theta_2|]$ , the diagonal resolution, and the earring closure through the corner folds—each step checked against the corresponding figure and formula of [9, 5]; the details and the validation batteries are in the source. The outcome is Figure 1, the pillowcase analogue of Smith’s Figure 31.

**2.3. The validation gate.** The purpose of the reconstruction is that it can be checked before it is used. Red and blue meet in nine points, which on  $P$  are one interior intersection near  $(1.61, 0.86)$  and two clusters of four near the corner column  $\gamma = \pi$ , at  $\theta \approx 1.52$  and  $\theta \approx 4.72$ ; this is Smith’s count of generators. A winding-number search (Section 3) finds exactly two immersed bigons, with all four vertices distinct, so the Floer differential has rank 1 on each and

$$\text{rank } HF((\mathcal{R}^\natural(T_1), 0), (\mathcal{R}_t(T_2), 0)) = 9 - 2 \cdot 2 = 5.$$

This is Theorem 1.1(i); it agrees with [9, Thm. 5.9] in every entry (nine points, two bigons with distinct vertices, rank 5), and since  $\text{rank } I^\natural(P(-2, 3, 5)) = 7$  [9, computed from Khovanov homology and the Alexander polynomial], it re-proves Smith’s conclusion  $b_2 \neq 0$ . Everything past this point is new.

### 3. THE COMBINATORIAL FLOER PACKAGE

**3.1. Bigons by winding numbers.** For two immersed curves in a surface, the Floer differential counts immersed bigons, which de Silva–Robbin–Salamon detect combinatorially by a winding function [2]. We use the pillowcase version. Fix generators  $x, y$  (intersection points of red and blue). A candidate bigon from  $x$  to  $y$  is a loop  $\Gamma = \alpha \cdot \beta$  formed by an arc  $\alpha$  of red from  $x$  to  $y$  and an arc  $\beta$  of blue from  $y$  to  $x$ . Lift  $\Gamma$  to  $T^2$  and let  $w: T^2 \setminus \Gamma \rightarrow \mathbb{Z}$  be its winding (intersection) function. Then  $\Gamma$  bounds an immersed bigon iff

- $[\Gamma] = 0 \in H_1(T^2)$ ;
- $w \equiv 0$  at the four corners of  $P$  (the disk avoids the orbifold points, which also fixes the additive constant);

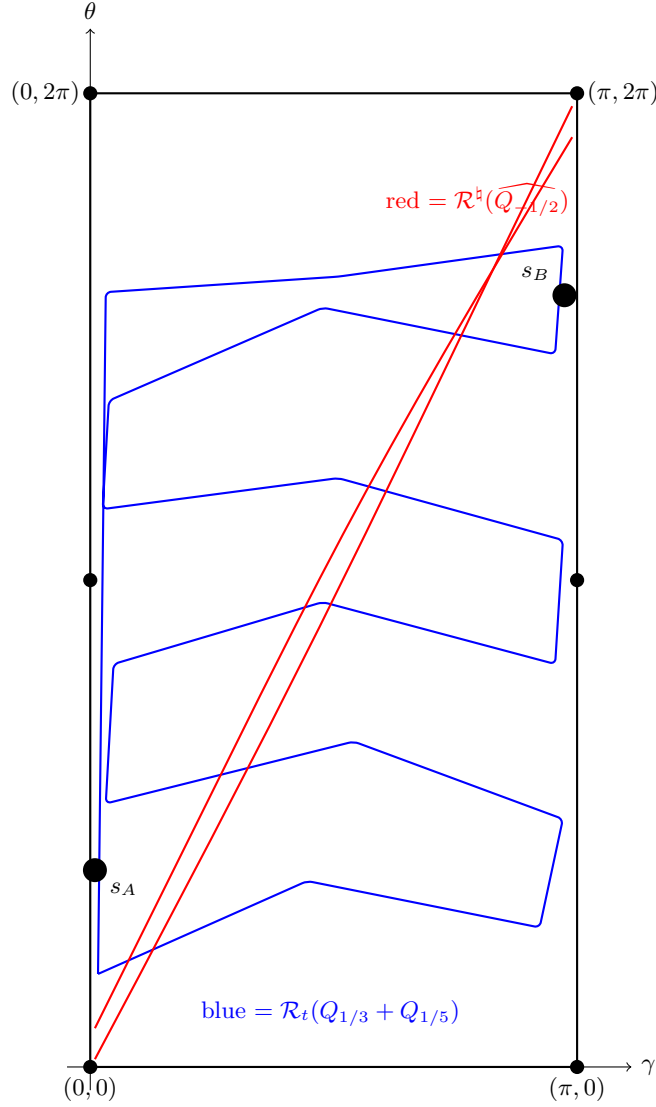


FIGURE 1. Schematic of the two Lagrangians for  $P(-2, 3, 5)$  on the fundamental domain  $[0, \pi] \times [0, 2\pi]$  of the pillowcase (Smith's Figure 31). Blue is the perturbed Conway sum  $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$ , an immersed circle with forty self-intersections clustered along the two seams  $\gamma \in \{0, \pi\}$ ; red is the earring figure-eight  $\mathcal{R}^\sharp(\widehat{Q_{-1/2}})$ . The two marked points  $s_A, s_B$  are the support of the computed bounding cochain  $b_2 = s_A + s_B$ ; they lie on opposite seams and are exchanged by the symmetry  $(\gamma, \theta) \mapsto (\pi - \gamma, -\theta)$ . The drawing is schematic; the counts below are exact.

- (c)  $w \geq 0$  on every complementary region; and
- (d) the two corners of  $\Gamma$  at  $x$  and  $y$  are convex, i.e. each is a single embedded sector of angle  $< \pi$  subtending winding jump  $+1$ .

The differential is  $\partial x = \sum_y (\#\{\text{bigons } x \rightarrow y\} \bmod 2) y$ , and  $\text{rank } HF = \#\{\text{generators}\} - 2 \text{rank } \partial$  over  $\mathbb{F}_2$ . Conditions (a)–(d) are exactly what our program tests, by adaptive sampling of  $w$  on both sides of every edge and a sector test at each corner; the delicate points (long edges under mod- $2\pi$  arithmetic, shallow-angle corners) are handled as recorded in the source.

**3.2. The lift bookkeeping.** Because we count on the double cover, we must be sure the cover count equals the pillowcase count. A pillowcase polygon lifts to exactly two  $\iota$ -related polygons on  $T^2$ ; fixing one lift of red (say the component  $C_+$  of the earring) selects exactly one lift of each pillowcase polygon whose red arc lies on  $C_+$ . Hence

$$\#\{\text{pillowcase polygons}\} = \#\{T^2 \text{ polygons with red arc on } C_+\},$$

and no division is needed. This is the same device by which the bigon count of §2.3 matches Smith's; it extends verbatim to the higher polygons below. A pillowcase self-intersection of blue has two  $T^2$  preimages, and a pillowcase polygon through it may close through either; we therefore take the union over preimages when tabulating a self-crossing's polygon contributions. *The one pitfall we flag:* four of the nine generators fold under  $T^2 \rightarrow P$  (they sit near  $\theta \approx 1.5$  on the cover but map near  $\theta \approx 4.7$  on  $P$ ), so proximity on  $P$  is *not* proximity on the cover; all locality tests must be phrased on the cover (by blue arc-length between anchor points), not in pillowcase coordinates. Ignoring this drops genuine polygons and was, in an intermediate version of this work, the source of a false negative that we correct here.

**3.3. The deformed differential.** Let  $\mathcal{R}$  be an immersed Lagrangian with self-intersections and  $b = \sum_i c_i s_i$  ( $c_i \in \mathbb{F}_2$ ) a bounding cochain on it. Following [1, §2.9] and [9, §5], the differential of the pair  $((\mathcal{R}^\natural(T_1), 0), (\mathcal{R}, b))$  counts immersed polygons two of whose vertices are generators (intersection points of red and blue) and each of whose remaining vertices is a self-intersection  $s_i$  in the support of  $b$ :

$$\partial_b(x) = \sum_y \left( \#\mathcal{B}(x, y) + \sum_i c_i \#\mathcal{T}(x, s_i, y) + \sum_{i < j} c_i c_j \#\mathcal{Q}(x, s_i, s_j, y) + \cdots \right) y \pmod{2}, \quad (2)$$

where  $\mathcal{B}$  are bigons,  $\mathcal{T}$  triangles (one  $b$ -vertex),  $\mathcal{Q}$  quadrilaterals (two  $b$ -vertices), and so on; these are the terms of the  $A_\infty$  maps  $\mu^1, \mu^2, \mu^3, \dots$  [9, §5.2]. Each polygon is a loop whose boundary alternates: an arc of red between the two generators, and arcs of blue between consecutive vertices, switching blue branch at each self-crossing. The very same winding conditions (a)–(d)—now with a convex-corner test at *every* vertex, at  $x$  and  $y$  (red  $\cap$  blue corners) and at each  $s_i$  (blue  $\cap$  blue corners)—decide whether the loop bounds an immersed disk. We validated the polygon counter on the unlink prototype of [9], where a single triangle through the earring pinch is the mechanism of that example.

The Maurer–Cartan equation (1) is the analogous count with *all* vertices at self-intersections: for each self-crossing  $s$ ,

$$\left[ \sum_k \mu^k(b, \dots, b) \right]_s = \#\mathcal{M}(s) + \sum_i c_i \#\mathcal{B}^\mathcal{R}(s_i, s) + \sum_{i, j} c_i c_j \#\mathcal{T}^\mathcal{R}(s_i, s_j, s) + \cdots = 0, \quad (3)$$

where  $\mathcal{M}$  are monogons (one-cornered teardrops, the curvature  $\mu^0$ ),  $\mathcal{B}^\mathcal{R}$  blue self-bigons ( $\mu^1$ ), and  $\mathcal{T}^\mathcal{R}$  blue triangles ( $\mu^2$ )—all polygons with boundary entirely on blue.

## 4. THE COMPUTATION

We now prove Theorem 1.1. Part (i) is the gate of §2.3. We take up (ii) the Maurer–Cartan side, then (iii) the rank side.

### 4.1. The Maurer–Cartan equation is vacuous to order two.

**Proposition 4.1.** *For the blue Lagrangian  $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$ , the self-Floer products  $\mu^0, \mu^1, \mu^2$  all vanish: there are no monogons, no self-bigons, and no self-triangles among its forty self-intersections. Consequently every  $\mathbb{F}_2$ -combination  $b$  of at most two self-intersections satisfies the Maurer–Cartan equation (1).*



*Proof (computation).* The three counts of (3) with all vertices on blue were enumerated by the winding-number test of §3.3, pruned on the cover by blue arc-length (§3.2). Monogons: for each of the forty self-crossings, the two blue arcs from one branch to the other were tested for a one-cornered immersed disk; none bounds. (The monogon counter was checked to *find* a genuine teardrop on a limaçon test curve, so the null result is not a false negative.) Self-bigons: for every ordered pair of self-crossings within a blue-arc window, the two blue arcs joining them were tested for a two-cornered disk; none bounds. Self-triangles: all  $\binom{40}{3}$  triples were tested (window-pruned); none bounds. Thus  $\mu^0 = \mu^1 = \mu^2 = 0$ .

Now let  $b = \sum_i c_i s_i$  with  $|\text{supp } b| \leq 2$ . In (3) the  $\mu^0$  term vanishes; the  $\mu^1$  term is a sum of self-bigon counts, which vanish; the  $\mu^2$  term is a sum of self-triangle counts, which vanish. The first potentially surviving term is  $\mu^3(b, b, b)$ , a count of blue quadrilaterals with three vertices in  $\text{supp } b$ ; but an immersed quadrilateral has four distinct vertices, so three of them cannot come from a two-element set. Hence  $\sum_k \mu^k(b, \dots, b) = 0$  identically for  $|\text{supp } b| \leq 2$ .  $\square$

Proposition 4.1 decouples the problem: for small support the Maurer–Cartan equation imposes *no* constraint, and the bounding cochain is pinned entirely by the requirement that the deformed homology have the correct rank. (That the self-products vanish is geometrically natural: blue is a taut resolution of the Conway sum, and its self-crossings are simple double points with no small enclosed disks. It also means  $b = 0$  is a bounding cochain—the naive theory is well defined—consistent with Smith computing rank 5 there.)

**4.2. The rank condition forces a bigon cancellation.** By §2.3 the undeformed differential  $\partial_0$  has rank 2, carried by the two bigons; call their generator pairs  $(x_1, y_0)$  and  $(x_4, y_6)$  in the labeling of Figure 1 (here  $y_0$  is the interior generator and  $(x_4, y_6)$  the pair in the  $\theta \approx 4.72$  cluster). We seek  $b_2$  with rank  $\partial_{b_2} = 1$ , so that rank  $HF = 9 - 2 = 7$ .

**Lemma 4.2.** *Over  $\mathbb{F}_2$ , a  $9 \times 9$  differential extending  $\partial_0$  by polygon terms as in (2) has rank 1 only if its polygon terms sum to cancel exactly one of the two bigon entries and introduce no other entry that raises the rank. In particular the deformed differential must equal a single nonzero matrix entry, up to the freedom of the surviving bigon’s row.*

*Proof.*  $\partial_0$  has its two nonzero entries in distinct rows and columns, hence rank 2. Adding an  $\mathbb{F}_2$ -matrix can lower the rank to 1 only by cancellation: the sum must have rank 1, i.e. all nonzero rows proportional. Since additions never remove the possibility of raising rank, the surgical outcome is the one in which the added polygon terms cancel exactly one bigon entry and add nothing that survives outside the remaining bigon’s row.  $\square$

Thus we must cancel a bigon: some polygon must contribute to the chain-level entry  $(x_4, y_6)$  or  $(x_1, y_0)$ . No triangle does (the single- $b$ -vertex terms of (2) were enumerated over all forty crossings; none has a triangle on either bigon pair). The cancellation is achieved at the next order.

**Proposition 4.3.** *There is an immersed quadrilateral with boundary vertices  $x_4, y_6$  and interior vertices the two self-crossings  $s_A, s_B$ ; it contributes to the entry  $(x_4, y_6)$  of (2). Neither  $s_A$  nor  $s_B$  carries any triangle, monogon, or any other quadrilateral with a second crossing in a minimal support. Hence for  $b_2 = s_A + s_B$ ,*

$$\partial_{b_2} = \partial_0 + \mathcal{Q}(x_4, s_A, s_B, y_6) \cdot E_{x_4, y_6}$$

*has the single entry  $(x_1, y_0)$ : the quadrilateral cancels the  $(x_4, y_6)$  bigon and nothing else changes. Therefore rank  $\partial_{b_2} = 1$  and rank  $HF = 7$ .*

*Proof (computation).* The quadrilateral is found by the polygon counter of §3.3: the loop  $\text{red}(x_4 \rightarrow y_6) \cdot \text{blue}(y_6 \rightarrow s_B) \cdot \text{blue}(s_B \rightarrow s_A) \cdot \text{blue}(s_A \rightarrow x_4)$ , switching blue branch at  $s_A$  and at  $s_B$ , satisfies

the winding conditions (a)–(d) with a convex corner at each of the four vertices; it is stable under enlarging the arc-length window over a wide range, so it is a genuine immersed disk and not a boundary artifact of the pruning. Enumerating the single-crossing (triangle) terms shows  $s_A$  and  $s_B$  carry none, and the full quadrilateral tabulation shows their only shared quadrilateral is the one above. Substituting into (2): the bigon terms give  $\partial_0$  (entries  $(x_1, y_0)$  and  $(x_4, y_6)$ ); the triangle terms vanish on  $\{s_A, s_B\}$ ; the single quadrilateral term adds  $E_{x_4, y_6}$ , canceling that bigon modulo 2. The result is the single entry  $(x_1, y_0)$ , of rank 1, so  $\text{rank } HF = 9 - 2 = 7$ .  $\square$

#### 4.3. Uniqueness and the proof of the theorem.

**Proposition 4.4.** *Among all bounding cochains supported on the forty self-crossings,  $b_2 = s_A + s_B$  is the unique one of minimal support with  $\text{rank } HF = 7$ : no single crossing achieves it, and  $s_A + s_B$  is the only two-crossing support that does.*

*Proof (computation).* Assembling (2) from the tabulated triangle and quadrilateral terms gives  $\partial_b$  as an explicit  $\mathbb{F}_2$ -polynomial in the coefficients  $c_i$ . A search over all supports by increasing size evaluates  $\text{rank } \partial_b$ : every single crossing gives  $\text{rank} \geq 2$  (indeed several raise it), and among all  $\binom{40}{2}$  pairs exactly one,  $\{s_A, s_B\}$ , gives rank 1. Every other pair either fails to touch a bigon or drags in a triangle that spoils the cancellation; for instance the geometrically nearest canceling pair (the two crossings flanking the  $(x_4, y_6)$  bigon) does cancel that bigon but its crossings carry triangles  $(x_3, y_6), (x_3, y_4), (y_6, y_5)$ , leaving rank 3. Minimality and uniqueness follow.  $\square$

*Proof of Theorem 1.1.* Part (i) is §2.3. Part (ii) is Proposition 4.1. For (iii), Proposition 4.3 exhibits  $b_2 = s_A + s_B$  with  $\text{rank } HF = 7$ , Proposition 4.1 shows it solves Maurer–Cartan, and Proposition 4.4 gives minimality and uniqueness. The two support crossings lie near  $(\gamma, \theta) = (0.03, 1.27)$  and  $(3.06, 4.98)$ , one on each seam, and are exchanged by  $(\gamma, \theta) \mapsto (\pi - \gamma, -\theta)$ .  $\square$

### 5. ROBUSTNESS, REPRODUCIBILITY, AND THE CONJECTURE

**5.1. Stability under perturbation.** The immersed curves depend on perturbation parameters (the shear resolving the seam circles, the earring amplitude and phase), and a genuine invariant must not depend on them. We ran the entire pipeline—assembly, gate, deformed differential, rank search, Maurer–Cartan check—at two independent choices of parameters. Both give nine generators, two bigons, rank 5; both give a *unique* two-crossing bounding cochain with  $\text{rank } HF = 7$ ; and the support crossings occur at the same pillowcase coordinates (within 0.02), on the same two seams. The value  $b_2 = s_A + s_B$  is stable.

**5.2. What is proved, computed, and conjectured.** The reconstruction of the Lagrangians and the deformed-differential formalism are re-implementations of [3, 4, 1, 5, 9], validated against [9] through the rank-5 gate. The Maurer–Cartan vanishing (Prop. 4.1), the canceling quadrilateral (Prop. 4.3), and the uniqueness (Prop. 4.4) are computer computations; the source and its validation batteries are included. The statement that  $b_2$  is *the* bounding cochain of Smith’s tangle—and hence that  $s_A + s_B$  is the correction that realizes the Atiyah–Floer counterpart of  $I^\natural(P(-2, 3, 5))$ —holds under the standing conjecture of [1] that the immersed-curve model computes the Floer-theoretic invariant, the same hypothesis under which [1, 9] are stated. Smith [9, Thm. 5.9] proves unconditionally that such a nonzero  $b_2$  exists; we compute its value within the model.

**5.3. Discussion.** Two features seem worth isolating. First, the correction here *removes* a differential (rank  $5 \rightarrow 7$ ), the opposite of the  $T(4, 5)$  correction, which *adds* one (rank  $9 \rightarrow 7$ ) [1]; canceling an existing bigon is more rigid than creating a new differential, because a canceling polygon must land on an occupied chain-level entry. That it is a quadrilateral rather than a triangle, and that its two vertices are otherwise polygon-free, is what makes the answer a



clean sum of two crossings. Second, the vanishing of  $\mu^0, \mu^1, \mu^2$  (Prop. 4.1) makes the Maurer–Cartan equation inert for small cochains: here the bounding cochain is selected purely by the rank matching, not by its own defining equation. Whether this decoupling persists for other decompositions—so that the computation of  $b$  reduces, as here, to a finite mod-2 rank search once the curves are in hand—is the natural question for turning the single example of Theorem 1.1 into a family computation, the open problem stated at the end of [10]. The pretzel knot  $P(-2, 3, 5) = T(3, 5)$  is the smallest  $n \equiv 5 \pmod{6}$  member of the  $(3, n)$  family, where the reduced complex first acquires the differential that on the pillowcase side is exactly this bounding cochain [10]; a uniform treatment of that family is the goal to which this example is the first data point.

## REFERENCES

- [1] G. Cazassus, C. Herald, P. Kirk, A. Kotelskiy, *The correspondence induced on the pillowcase by the earring tangle*, J. Topol. **15** (2022), 2472–2543. arXiv:2010.04320.
- [2] V. de Silva, J. Robbin, D. Salamon, *Combinatorial Floer homology*, Mem. Amer. Math. Soc. **230** (2014), no. 1080.
- [3] M. Hedden, C. Herald, P. Kirk, *The pillowcase and perturbations of traceless representations of knot groups*, Geom. Topol. **18** (2014), 211–287. arXiv:1301.0164.
- [4] M. Hedden, C. Herald, P. Kirk, *The pillowcase and traceless representations of knot groups II: a Lagrangian–Floer theory in the pillowcase*, J. Symplectic Geom. **16** (2018), 721–815. arXiv:1501.00028.
- [5] C. Herald, P. Kirk, *An endomorphism on immersed curves in the pillowcase*, arXiv:2407.11247.
- [6] P. Kronheimer, T. Mrowka, *Knot homology groups from instantons*, J. Topol. **4** (2011), 835–918. arXiv:0806.1053.
- [7] P. Kronheimer, T. Mrowka, *Khovanov homology is an unknot-detector*, Publ. Math. IHÉS **113** (2011), 97–208. arXiv:1005.4346.
- [8] X.-S. Lin, *A knot invariant via representation spaces*, J. Differential Geom. **35** (1992), 337–357.
- [9] K. Smith, *Perturbed traceless  $SU(2)$  character varieties of tangle sums*, arXiv:2412.06066.
- [10] B. J. Wuebben, *Trace-free  $SU(2)$  characters and  $\mathbb{Z}/4$  instanton gradings for two-bridge and  $(3, n)$ -torus knots*, companion paper (2026).

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